Mengyuan (Cecilia) Xue

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Education

The University of Texas Southwestern Medical Center, Dallas, TX *08/2025 – 05/2030 (Expected)*
Ph.D. in Biomedical Engineering
- Research track in Biomedical & Molecular Imaging

Johns Hopkins University, Baltimore, MD *08/2021 – 08/2022*
M.S.E. in Biomedical Engineering
- Research track in Imaging and Medical Devices
- GPA: 3.95/4.0

Washington University in St. Louis, St. Louis, MO *08/2017 – 05/2021*
B.S. in Biomedical Engineering
- Double major in Financial Engineering, minor in Math and Computer Science
- Honor Society Membership: Tau Beta Pi (Engineering), Alpha Eta Mu Beta (BME), BMES Student Member
- GPA: 3.93/4.0 (Summa cum laude, Dean's List all semesters)

Papers & Conference Presentations

1. **[Journal]** **Mengyuan Xue***, Diego Gonzalez*, Emmanuel Ospika, Xue Gao, Peter B. Lillehoj, "Rapid and automated interpretation of CRISPR-Cas13-based lateral flow assay test results using machine learning", *Sensors & Diagnostics*, 2025, https://doi.org/10.1039/D4SD00314D
2. **[Conference]** **Mengyuan Xue***, Bochun Mei*, "Joint Supervised and Self-Supervised Training with Acquisition-Robust Techniques for Accelerated 4D Flow MRI Reconstruction", *29th International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI 2026-CMRx4DFlow2026)*, (Sep 2026), https://arxiv.org/pdf/2609.38644
3. **[Conference]** **Mengyuan Xue**, Ruchika Mitbander, Jennifer Carns, Richard A. Schwarz, Nadarajah Vigneswaran, Ann Gillenwater, Rebecca Richards-Kortum, "A deep learning mucosal segmentation algorithm for a mobile Detection of Oral Cancer (mDOC) platform", Proc. SPIE 13186, *SPIE Translational Biophotonics + Additive Manufacturing for Photonics 2024*, (23 Sep 2024), https://doi.org/10.1117/12.3034638
4. **[Conference]** **Mengyuan Xue***, Diego Gonzalez*, Weinan Wang, Emmanuel Ospika, Xue Gao, Peter B. Lillehoj, "Development of a Deep Learning Model for Accurate Interpretation of CRISPR-Cas-based Lateral Flow Assay Test Results", *The Ken Kennedy Institute AI In Health Conference*, 2024
5. **[Conference]** **Mengyuan Xue**, Yameng Gu, Yusheng Li, Lingzhi Hu, Xishan Sun, "Attenuation Correction of fNIRS Optic Fibers for PET Reconstruction on uPMR790 System", *InterfaceRice*, 2024
6. **[Conference]** **Mengyuan Xue**, Yameng Gu, Yusheng Li, Lingzhi Hu, Xishan Sun, "Attenuation Correction of fNIRS Optic Fibers for PET Reconstruction on uPMR790 System", *The International Society for Magnetic Resonance in Medicine (ISMRM) Workshop on PET/MRI*, 2023

(* indicates equal contribution)

Research Experience

Rice University *Houston, TX*
Research Specialist | Lillehoj Research Group *08/2023 – present*
Project: Development of rapid approach for interpreting CRISPR-Cas13-based lateral flow assay results.
- Collected 350+ high-resolution images of 12 lateral flow devices with mobile phones from multiple sites. Adopted a random sampling strategy to crop images, creating an augmented dataset of 3,500+ negative and 4,500+ positive lateral flow assay images.
- Designed and implemented three neural network architectures (U-Net-based, MobileNet-based), achieving 95% segmentation accuracy on test set across varying resolutions, contrast and lighting conditions.
- Developed *ClassNet*, a lightweight classification network to interpret binary assay results from lateral flow strips, achieving 97.7% accuracy on the test set (1100+ negative/1300+ positive), with sensitivity > 98.2% and specificity > 96.5%.

Research Specialist | Optical Spectroscopy and Imaging Lab *12/2023 – 04/2025*
Project: Development of end-to-end workflow for early detection of oral cancer on mobile-based platforms.
- Pre-processed and prepared datasets from raw clinical data, including ~430 autofluorescence and ~700 white light oral mucosa images.

- Designed and implemented five neural network architectures as the backbone of segmentation models, achieving segmentation Dice scores exceeding 0.8 on the test set for 84% of autofluorescence (~130 images) and 75% of white light images (~220 images).
- Optimized a classification model with MobileNet-V3 as the backbone and used ensemble learning to predict referral decisions from masked autofluorescence and white light images, achieving sensitivity > 90%, and specificity >80%.
- Developed and deployed an Android application that integrated the models, enabling a fully real-time workflow with referral decisions generated within 10 minutes of imaging sessions, validated through proof-of-concept experiments on normal volunteers.
- **Miscellaneous:** Assisted in assembling HRME (High-Resolution Micro Endoscopy) from separate parts.

United Imaging *Houston, TX*
R&D Intern | Future Lab *05/2023 – 11/2023*
Project: Attenuation correction of fNIRS optic fibers for PET reconstruction.
- Developed an image processing pipeline for correcting attenuation from optical fibers of a PET/MR/fNIRS multi-modal imaging system during PET reconstruction.
- Tested the algorithm on human data from 10 volunteers and achieved a 90% reduction in homogeneity deviation on a barrel source phantom with fNIRS optic fibers during PET reconstruction.
- Implemented the algorithm into a C++ engineering version for use in the PET/MR uPMR790 system.
- First-authored an abstract accepted at the ISMRM Workshop on PET/MRI and co-authored a journal paper currently in preparation.

Johns Hopkins University *Baltimore, MD*
Graduate Research Assistant | Advanced Imaging Algorithms and Instrumentation Lab *02/2022 – 05/2023*
Project 1: Predicting and Recovering Wavelet- and Fourier-Domain Radiomics Under Different Resolution Situations
- Simulated blur and noise with a cascaded system and analysis model, generating 170+ combinations of 2D modulation transfer function (MTF) and point spread function (PSF) with realistic blur/noise properties.
- Compiled and validated experimental data, evaluating performance of the framework on 34 radiomic feature values.
- Optimized the deconvolution process, using a radiomics-based and distance-based objective function to improve the framework's accuracy.
- Extended the radiomics recovery framework through the derivation of an analytical probability density function and linear image transformations.

Project 2: Reproducibility analysis on 3D-printed and simulated phantoms
- Developed a pipeline for comparing physical 3D-printed phantoms with simulations and conducted reproducibility studies through CT data reconstruction, 3D-3D image registration and in-depth spectral analysis.
- Simulated five 3D phantoms based on power law relationship, printed 12 phantoms using FDM 3D-printers.
- Scanned the physical phantoms with in-lab X-ray imaging bench and conducted reconstruction from the projection data.

Miscellaneous:
- Optimized *Pyradiomics*, an open-source radiomics package, to incorporate custom radiomics matrices and axes.
- Developed a proof-of-concept GUI application for visualizing overlapping CT images, ROIs, and spectral CT material maps.

Philips Healthcare *Remote*
R&D Intern (remote) | Division of Radiation Oncology *05/2021 – 10/2021*
- Organized and summarized training materials on CT/MRI radiotherapy and MR-sim from UMC Utrecht for internal use.
- Delivered training on image-guided radiotherapy and its clinical applications to division members and engineers.
- Summarized the Dual-energy CT AAPM task report into presentations and shared key insights with the division.
- Gained fundamental knowledge in the physics of CT/MRI and the applications in radiotherapy and clinical settings.
- Edited and translated multiple manuscripts intended for publication, ensuring clarity and accuracy.

Washington University in St. Louis *St. Louis, MO*
Undergraduate Research Assistant *05/2020 – 12/2020*
Project: Development of vessel segmentation pipeline on mouse brain photoacoustic images
- Prepared training, testing, and validation data, creating binary labels using a thresholding method, and preprocessed and augmented data/labels for U-Net-based vessel segmentation.
- Modified U-Net parameters and trained the neural network using high-quality computer-simulated data, applying transfer learning to original vessel data, which significantly improved overall training accuracy.

- Developed an automated vessel segmentation algorithm by integrating image processing techniques, edge detection, and vessel segmentation, and systematically labeled common vessels across a series of time-lapsed brain images.

Teaching & Mentoring

Johns Hopkins University — *Baltimore, MD*
Graduate Teaching Assistant — *08/2022 – 05/2023*

- Courses TA'ed:
  - 580.475/477 - Biomedical Data Science Laboratory | Instructor: Dr. Brian Caffo
  - 585.601 - Physiology for Applied Biomedical Engineering I | Instructor: Dr. Caitlin Torgerson
- Held weekly office hours to help students understand course material and resolve coding issues for assignments.
- Led weekly lab sessions, providing in-depth reviews of key concepts from class and reinforcing learning through practical application.
- Took a lead role in clarifying complex concepts, ensuring students grasped essential topics discussed during lectures.

Washington University in St. Louis — *St. Louis, MO*
Undergraduate Teaching Assistant — *08/2019 – 05/2021*

- Courses TA'ed:
  - BME 140: Introduction to Biomedical Engineering – Fall 2020
  - BME 320: Bioengineering Thermodynamics – Fall 2019
  - CSE 131: Introduction to Computer Science – Fall 2019
  - CSE 217: Introduction to Data Science – Spring 2021
  - CSE 247: Data Structures and Algorithms – Fall 2020
  - CSE 417: Introduction to Machine Learning – Spring 2021
- Led lab sessions for MATLAB and CAD practice to help students apply course material through hands-on projects.
- Graded assignments, lab reports, and exams, while holding weekly office hours to provide additional support with coding homework, lab assignments, worksheets, and exam preparation.

Engineering Student Tutor — *08/2018 – 05/2021*

- Tutored 20+ engineering students over three years, dedicating 7-10 hours weekly to improve their understanding of engineering concepts and enhance their academic performance.
- Developed and organized preparation materials to help students excel in their courses, ensuring a tailored approach to individual learning needs.

Grader — *08/2018 – 05/2021*

- Courses Graded:
  - Math 217: Differential Equations – Fall 2018, Spring 2019 (Lead Grader)
  - Math 310: Foundations for Higher Mathematics – Fall 2019, Spring 2020 (Lead Grader)
  - Math 308: Mathematics in Physical Science – Spring 2020
  - Math 493: Probability – Spring 2021

Skills

Biomedical Engineering
Real-Time PCR | Assay Development | Animal Studies | Medical Imaging | Image Reconstruction | Data Acquisition | CAD | Mathematical Modeling | Image Processing | Experimental Design | Public Speaking | Scientific Writing

Programming & Software
MATLAB | Python | Java | C++ | Kotlin | Cadence Virtuoso | LaTeX | Machine Learning | Deep Learning | Modeling & Simulations | Arduino | Android Studio

Selected Coursework

Washington University in St. Louis
- Quantitative Physiology
- Introduction to Machine Learning
- Introduction to Artificial Intelligence
- Mathematics of Imaging Science
- Biological Imaging Technology
- Fundamentals and Applications of Modern Optical Imaging

Johns Hopkins University
- Radiology for Engineers
- Introduction to Computational Medicine: Imaging
- Modern Optical Microscopy
- Principle and Application of Modern X-Ray & Tomography
- Systems Pharmacology and Personalized Medicine